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Quantifying the impact of extreme weather on China's hydropower–wind–solar renewable energy system

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Carbon reduction goals have driven China to become the world's largest renewable energy system (RES) that is dominated by hydropower, wind power and solar power. However, the meteorological sensitivity of wind and solar power greatly affects the reliability and generating capability of the RES, particularly in extreme weather events. Quantifying the electricity supply and flexibility of hydropower is crucial for compensating extreme wind and solar power generation. Here we investigate the influence of extreme weather combinations and future climate on the generating capability of the national RES and quantify the flexibility demand and hydropower supply in typical extreme weathers. Our analysis reveals that the annual utilization hours of the hydropower–wind–solar system are projected to decline by nearly 12% from the current stage to 2060 under conditions of extreme drought, low wind and weak solar radiation. When encountered with extremely strong solar radiation and wind, the probability of flexibility shortages in hydropower is estimated to rise to 47% by 2030 and further increase to 60% by 2060. Nearly half of the provinces will require tens of millions of kilowatts of energy storage by 2030 to supplement the flexibility supply gap of hydropower, and by 2060, both the number of provinces that require such a large-scale energy storage and the storage capacity needs might double. Our findings provide early warnings in extreme electricity supply and underscore the growing necessity for building dispatchable power plants and exploring the flexibility of existing hydropower systems.

Increasingly severe climate conditions and energy and environment crises inspire many countries to set carbon reduction targets for sustainable development[1–3]. A unified consensus worldwide is formed to develop renewable energy[4]. China has implemented this approach to achieve ambitious carbon-neutrality goals[5–7]. By the end of 2023, the installed capacity of hydropower, wind power and solar power has reached 422 GW, 441 GW and 609 GW (ref. 8), respectively, all ranking first in the world. These renewable energy sources have become the dominant power source of the national electricity system, and their share in the total installed capacity will increase from the current 50.4% to about 88% by 2060 (ref. 9). In this context, the uncertainty and intermittency of wind and solar power pose large operational risks to electrical grids, such as power shortage and curtailment. Hydropower can play a critical role in integrating large-scale wind and solar power owing to strong dispatchable flexibility across seasonable, monthly, weekly, daily and hourly timescales. Therefore, the integrated operation of hydropower, wind power and solar power has been widely recognized and utilized to overcome the common issue.

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However, hydropower, wind power and solar power depend largely on meteorological conditions such as wind speed, solar radiation and rainfall[10,11]. Severe and extreme weathers bring enormous power supply pressure and operational risks to the national power system[12,13]. In recent years, there have been frequent power system warnings and risk events caused by extreme weather, such as continuous weak winds in Northeast China (one of the three major new energy regions)[14] in 2021, hydropower droughts in Sichuan (the largest hydropower province)[15] in 2022 and Brazil (the second largest country in terms of installed hydropower capacity)[16] in 2021 and extremely high temperature in South Australia[17] in 2017. Large-scale unplanned power shortages or curtailments have been caused. As highly renewable shares continue to rise rapidly in the world, the likelihood of similar power safety incidents is increasing along with frequent occurrences of extreme weather events. There is an urgent need to quantify the impacts of extreme weather and future climate change on the operations of hybrid systems (including hydropower, wind power and solar power) and explore the contributions of hydropower in mitigating adverse impacts and providing power flexibility for the integration of variable renewable power sources.

Such prominent issues have attracted increasing attention from researchers. Some works found that extreme weather causes extreme electricity demand, such as extreme cooling and heating demand in California and Texas in 2020 and 2021, respectively[18,19]. Another finding revealed that extreme weather significantly damages power facilities[20,21] or affects the electricity supply[22–25]. Other researchers tried to quantify the power supply capability of renewable energy for future power systems and predict the potential impact of power shortages[26,27]. These studies focus mainly on a certain aspect of variable renewable power sources under extreme weather conditions, but the role of hydropower in supporting large-scale wind and solar power has received little attention. In fact, numerous large hydropower stations in a hybrid system can play a critical role as flexible power sources. This highlights the need for systematic research on generating capability, flexibility demand, hydropower flexibility supply and power balance of the hybrid system in extreme weather events. Such a study is important for planning renewable power sources, assisting in identifying and addressing operational risks, and strengthening the flexibility role of hydropower.

This study focuses on the impact of extreme weather conditions and future climate changes on China's renewable system operations. The research consists of four main parts. The first describes extreme weather conditions and their combinations of different extreme scenarios on multiple types of power sources in the country with highly uneven resource distribution. The second evaluates the electricity supply of the hybrid system and the power shortages in provincial power grids under different weather scenarios at present, by 2030 and by 2060. The third quantifies the flexibility demand brought by integrating variable renewable power into electricity networks. The fourth evaluates the relationship between flexibility demand and supply of provincial system with various types of power sources and optimizes the capacities and durations of energy storage required in 2030 and 2060 for different provinces.

Compared with previous research works, the main contributions of this study are as follows. (1) The wind and solar capacity factors are proposed to describe the abundance of overall national resources. These two indicators simultaneously consider the uneven distribution of resources and the degree of resource development in different regions. (2) An integrated optimization is implemented to main hydropower plants in large rivers of the country. Correspondingly, an optimization model is developed to enhance the average generating capability of hydropower plants in the dry season and, thus, guarantee monthly electricity supply and dispatchable flexibility of provincial power grids over the entire time horizon. (3) A complete quantitative method is developed to analyse the generating capability, flexibility demand and supply of billion-kilowatt-scale national systems with uneven resource distribution in various extreme weather conditions. (4) This study proposes a probability-based evaluation method for flexibility demand and supply of all power sources in provincial power grids of China. This method is useful to enhance the accuracy of the analysis of long-term power shortage and surplus under extreme weather conditions. In addition, our findings can provide early warnings in extreme electricity supply and underscore the growing necessity for building dispatchable power plants and exploring the flexibility of existing hydropower systems.

Continuous extreme weather descriptions

This section will analyse the distribution of China's hydropower, wind and solar resources and their change characteristics to generate nationwide extreme weather scenarios. Long series of historical inflows and meteorological data over 40 years are adopted here.

Resource distribution

China is endowed with abundant hydropower, wind and solar resources, but their distribution is extremely uneven. As shown in Fig. 1, the hydropower is concentrated mainly in the southwest region, accounting for more than 70% of the country's total hydropower resources. We divide the southwest region into eight main watersheds: Jinsha River, upper Yangtze River, Yalong River, Lancang River, Dadu River, Red River, Nanpan River and Wujiang River. The hydropower abundance of each watershed is evaluated by using the inflow of a representative hydropower plant over decades. Similarly, China's wind and solar resources also exhibit significant regional characteristics. To quantify the resource abundance, we develop a wind capacity factor and a solar capacity factor. The weighted averages of the corresponding capacity factors of all provinces are used to represent the national level of wind or solar resources.

Resource gradation

To clarify the interannual changes in energy resources, we classified four decades of historical records or calculation values (1980–2019) into five levels (extremely dry/weak, dry/weak, normal, wet/strong and extremely wet/strong) based on inflow, wind capacity factor and solar capacity factor. The specific methods are given in the 'Description of extreme weather' section in the Methods.

Figure 2a shows the inflows of major river basins from 1980 to 2019 in China. There exist significant variations in the abundance of hydropower resources on an annual basis among river basins. The annual inflow ranges from 17,870 $m^3 s^{-1}$ in the driest year up to 29,977 $m^3 s^{-1}$ in the wettest year, with a gap of 67.7%. Two significant situations can be observed, that is, continuous extremely wet years from 1998 to 2001 with an average inflow of 27,851 $m^3 s^{-1}$, and continuous extremely dry years from 2009 to 2011 with an average inflow of 20,916 $m^3 s^{-1}$. Figure 2b gives the change profile of wind capacity factors over 40 years. A large interannual variation of wind resources with a difference of 14.3% between the maximum and minimum indexes is found. Continuous extremely strong-wind and weak-wind years occurred from 1987 to 1988 and 1982 to 1986, respectively. The corresponding wind capacity factors of the two cases are 0.260 and 0.242, with a gap of 7.6%. Figure 2c lists solar capacity factors over decades in the country. The solar capacity factors vary between 0.134 and 0.144 over 40 years from 1980 to 2019, with a maximum difference of 7.8%. Deviations from the long-term mean solar capacity factors range from −3.5% to 3.9%. The interannual variability of solar capacity factors is thus lower than that of wind, which experienced deviations of −5.0% to 8.7%. Similarly, continuous extremely strong-solar radiation and weak-solar radiation years from 2004 to 2009 and 1989 to 1991 are shown in Fig. 2c.

Combinations of extreme weather

Supplementary Fig. 1 shows comprehensive gradations of hydropower, wind and solar resources from 1980 to 2019. The occurrence of

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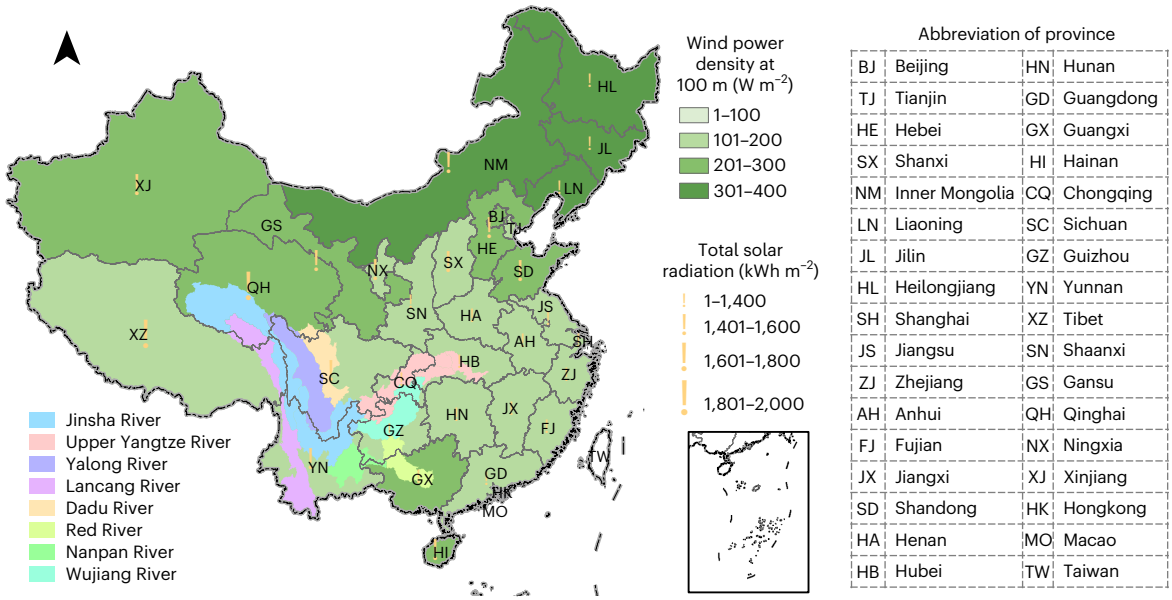


Fig. 1 | Hydropower, wind and solar resource distribution. The wind power density and total solar radiation are obtained from China Wind and Solar Energy Resources Bulletin (2023) issued by CMA Wind and Solar Energy Centre. Hydropower resources are distributed on the basis of watersheds. Wind and solar resources are distributed on the basis of provinces.

wet/dry and strong/weak years shows some regularity. Because the large inflow is beneficial for the power supply, we focus on the impact of continuous drought.

In dry years, there are two types of combination for wind and solar radiation. One is years with weak wind and strong solar radiation, such as 1986, 1997, 2007, 2011 and 2019; the other is years with strong wind and strong solar radiation, such as 2009, 2010 and 2013.

The continuous extreme weather greatly threatens the power supply, especially the flexible power of electrical systems. Based on historical weather events, we determined three typical extreme weather conditions: (1) scenario 1: continuous drought, weak wind and solar radiation; (2) scenario 2: continuous drought, weak wind and strong solar radiation; (3) scenario 3: continuous drought, strong wind and solar radiation. Considering the impacts of future climate change, we have also added two scenarios, SSP126 (scenario 4) and SSP585 (scenario 5) for analysis.

Analysis of generating capability

This section focuses on analysing the generating capability of national hybrid system and power shortage of provincial power grids under extreme weathers in 2023, by 2030 and by 2060. The installed capacities of various types of power source are shown in Supplementary Fig. 2. The installed capacities of current stage utilize actual data at the end of 2023, and the values of the other two stages come from research on China's 2030 energy and electricity development plan and outlook for 2060 (ref. 9).

We evaluate the national generating capability of hydropower, wind power and solar power under different extreme weather conditions (see the 'Evaluation method for generating capability' section in the Methods). In hydropower operations, inadequate power supply is prone to occur in extremely dry years, especially during the dry season. Therefore, an optimization model for maximizing the minimum monthly generation is used to determine the monthly hydropower generation, intending to improve the power supply capability during the dry season. Figure 3a shows monthly hydropower generation profiles in extremely dry years. The total annual generation is 1,234 TWh, with a decrease of 17.5% compared with the annual average value (average annual utilization hours from 2018 to 2023 multiplied by current installed capacity). Moreover, the power generation during the dry season (from November to next May) is 610 TWh, accounting for 49.4% of the annual generation, which is beneficial for the stability of the power supply.

Wind and solar power generation heavily depends on weather conditions, and it is hard to dispatch them as needed. Thus, their generating capabilities are directly evaluated using capacity factors and installed capacities. Figure 3b shows monthly generation profiles of wind power in two types of extreme years. During peak months (from November to next April), the generation in the extremely strong-wind year is greater than that in the extremely weak-wind year. The reason lies in that the wind speed in the provinces with large wind power installed capacity (such as Inner Mongolia (NM), Gansu (GS) and Hebei (HE)) is quite high in the spring under the influence of Siberia–Mongolia high pressure. This implies that surplus wind electricity in the extremely strong-wind year is produced mainly in the peak season, imposing a huge pressure on the consumption of wind power. The monthly solar generation profiles in two types of extreme years are shown in Fig. 3c. There is a significant increase in power generation during the first and second quarters in the extremely strong-solar radiation year compared with values of the same periods in the extremely weak-solar radiation year, whereas similar generations are found in other months. This finding implies that integrating solar power from January to June should be the focus. Furthermore, the gap between wind generation production in the extremely strong-wind and weak-wind years is 11.0%, while the gap in solar power is 7.4%. In summary, extreme weather has a greater impact on the generating capability of wind power than that of solar power.

Figure 3d compares the total generation of the hybrid system under different extreme scenarios in 2023. The annual power generation of the three extreme scenarios is 2,872 TWh, 2,925 TWh and 3,026 TWh, respectively. The increasing rate from scenario 1 to scenario 2 is 1.8%, while the rate from scenario 2 to scenario 3 is 3.5%. This results from the greater impact of change in wind resources on the generating capability, which has been verified in the comparison between Fig. 3b and Fig. 3c. Furthermore, the difference in monthly generation between scenario 1 and scenario 2 is reflected in January to June, which is consistent with the results of solar power in Fig. 3c. Correspondingly, the difference between scenario 2 and scenario 3 concentrates in November to next January, like the results of wind power in Fig. 3b.

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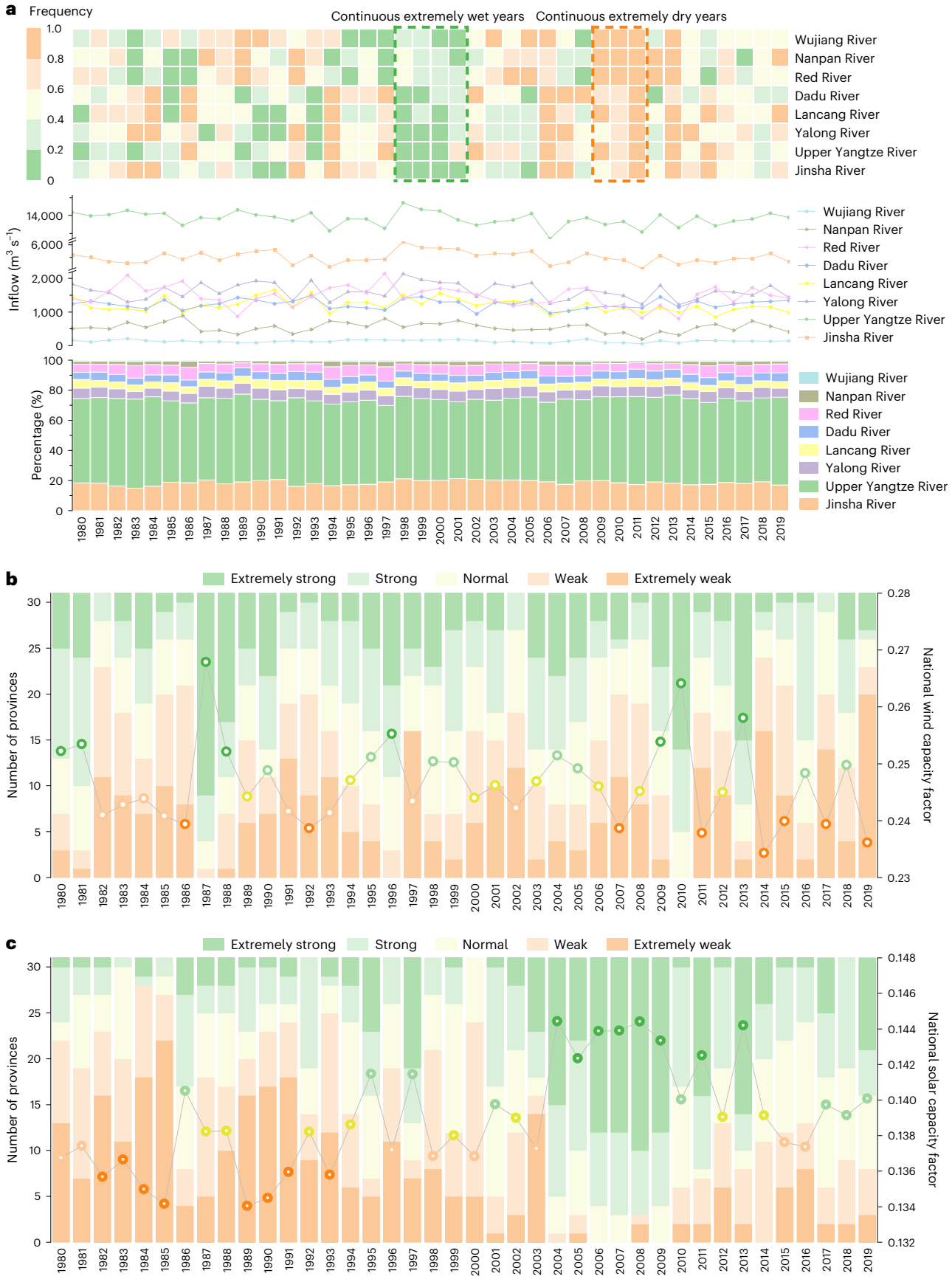


Fig. 2 | The abundance of hydropower, wind and solar resources. a, Inflows of major river basins from 1980 to 2019 in China. **b**, Wind capacity factors in 40 years. The histogram shows the number of provinces with different levels of wind capacity factors, and the line shows the national wind capacity factors. **c**, Solar capacity factors in 40 years. The histogram shows the number of provinces with different levels of solar capacity factors, and the line shows the national solar capacity factors.

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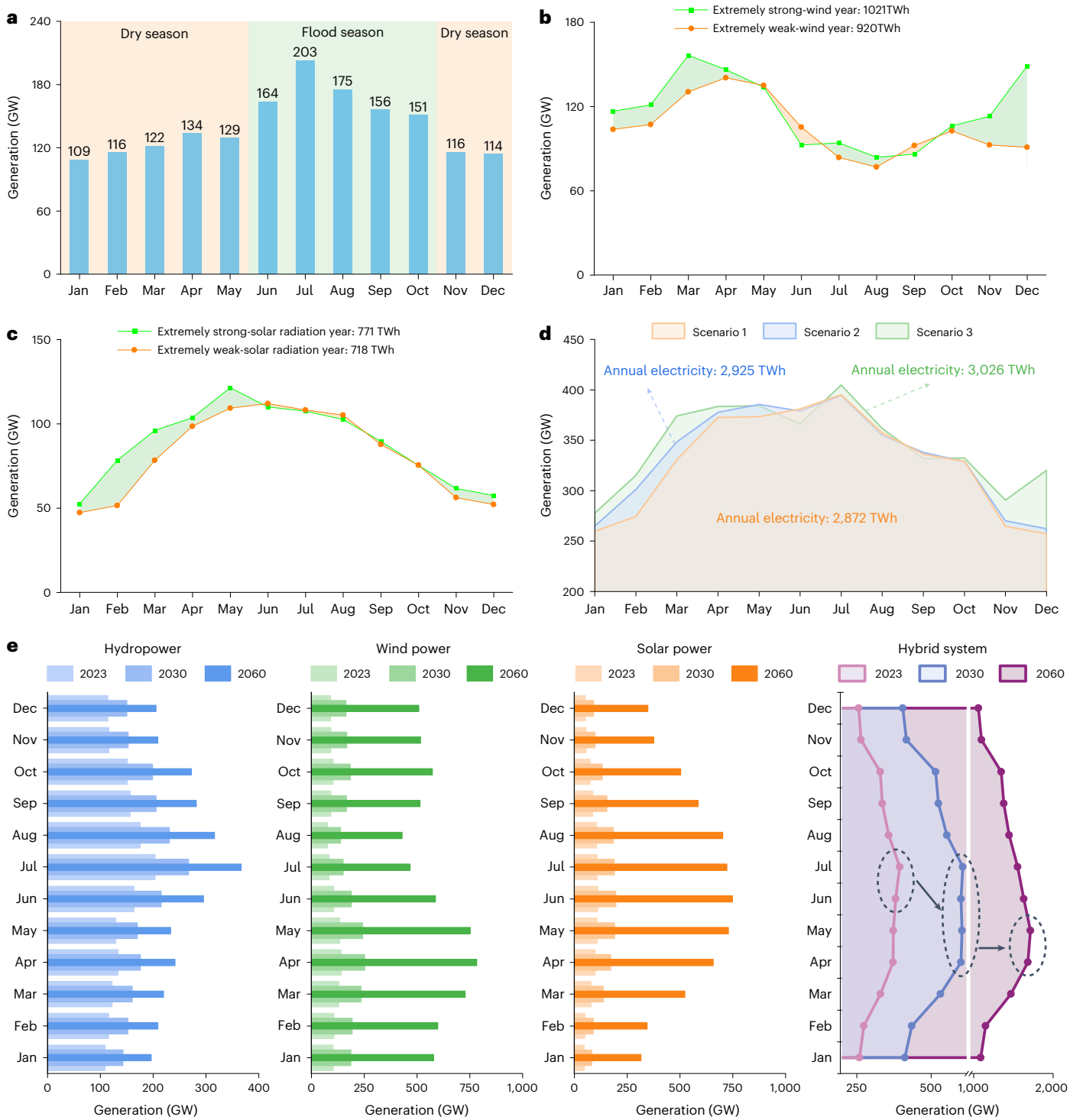


Fig. 3 | The generating capability of hydropower, wind power and solar power. **a**, The generating capability of hydropower in 2023. **b**, The generating capability of wind power in 2023. **c**, The generating capability of solar power in 2023. **d**, The generating capability of the hybrid system in 2023. **e**, The generating capability in 2023, 2030 and 2060 of scenario 1. Note that the generating capabilities of Hongkong (HK), Macao (MO) and Taiwan (TW) are not included in the study owing to a lack of data collection for these areas.

The generating capability of the hybrid system will change greatly as the variable renewable energy (VRE; wind and solar power) increases in the future. Figure 3e displays the total generation in different stages. Comparing the results of different stages, the peak generation of hybrid system gradually moves from June and July to April and May over time. The main cause is that the generation of hydropower concentrates from June to July (Fig. 3a), while strong wind power usually occurs from April to May (Fig. 3b). The variation of power capacity structure results in a high risk of power shortages from June to July. Moreover, we find that the growth rate of power generation in the hybrid system is slower than that of installed capacity. This is because the utilization hours of wind and solar power that contribute the most to the newly added capacity are much fewer than those of hydropower. Under the same capacity conditions, the generating capability of hydropower is about 1.5 times that of wind power and 2.5 times that of solar power.

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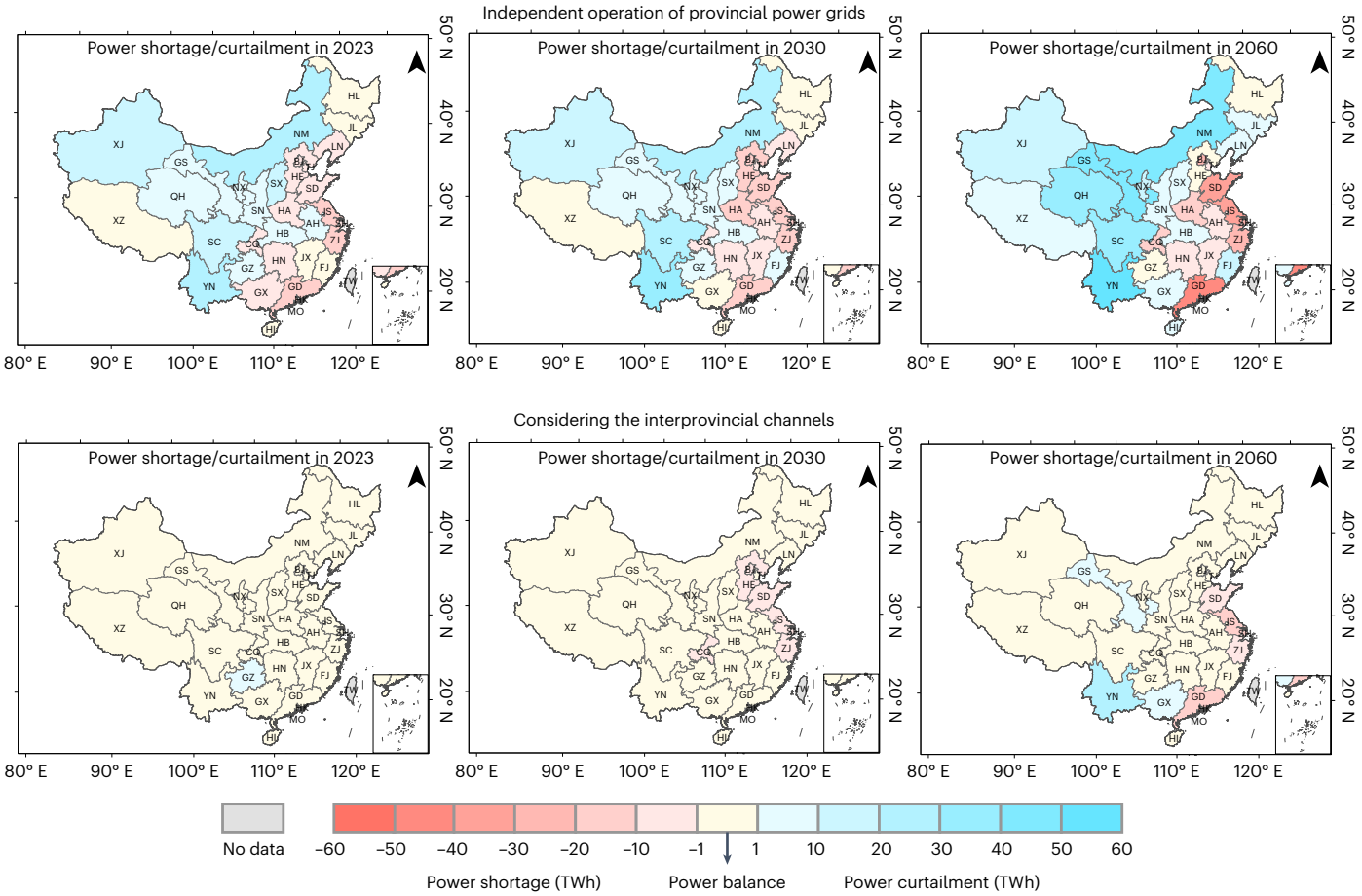


Fig. 4 | Distribution of power shortages and curtailments for each province in July in scenario 1. Top: the three graphs display the distribution of power shortages and curtailments under independent operations of provincial power grids. Bottom: the three graphs display the distribution of power shortages and curtailments considering interprovincial transmission channels.

The annual generating capabilities for wind and solar power under different extreme scenarios are given in Supplementary Table 1. The maximum annual generation of wind power occurs in the extremely strong-wind scenario in 2030 and SSP585 scenario in 2060. The maximum annual solar power generation occurs in the SSP126 scenario in 2030 and extremely strong-solar radiation scenario in 2060, while the minimum annual generation in two future stages occurs in the extremely weak scenario. We find that the extremely strong and weak scenarios are even more unfavourable than SSP126 and SSP585.

The monthly generating capability of wind and solar power under different scenarios in 2030 and 2060 are shown in Supplementary Fig. 3. In 2030, the difference between the maximum and minimum monthly generation of wind power under SSP126 and SSP585 is 63 GW and 89 GW, respectively. They are much smaller than those under the extremely strong-wind scenario (131 GW) and extremely weak-wind scenario (114 GW). Similar results occur in 2060. As a result, the electrical grids would be more vulnerable to power shortages and curtailments under extremely strong-wind and weak-wind scenarios. Different from wind generation profiles, the monthly generation profiles of solar power under different weather scenarios are relatively consistent. The overall generation output from November to February of the following year is low, while it is relatively high from April to August. However, the negative impact caused by such extreme solar power situation might be weakened, as there is a clear complementarity between solar and wind power generation.

The uneven spatial distribution of hydropower, wind power and solar power sources (Supplementary Fig. 4) holds vast potential for exploring the interprovincial power complementarity by considering power transmission. Therefore, interprovincial power exchanges are further made to analyse the changes in power shortages and curtailments.

Here, we take a typical scenario (July in scenario 1) to analyse the spatial distribution of power shortages and curtailments. Figure 4 shows comparisons with and without interprovincial power exchange. Under independent operations, most provinces in western China generate more electricity than their own power demand in scenario 1, resulting in an electricity surplus. By contrast, there exist common power shortages in eastern provinces. The top graphs in Fig. 4 indicate this situation. Using interprovincial power transmission channels, most provinces achieve the power balance. However, there are still several provinces with an electricity surplus and additional provinces with power shortages in 2030 and 2060. This is due to transmission capacity limitations. Thus, there might be a need for expanding the interprovincial transmission capacity in the future, especially between China Southern Power Grid (including Yunnan (YN), Guizhou (GZ), Guangxi (GX), Guangdong (GD) and Hainan (HI)) and State Grid Corporation of China (including the other provincial power grids).

The national power shortages and curtailments under different scenarios are depicted in Supplementary Fig. 5. There is a large electricity gap in 2030. The maximum power shortage is 424 TWh in scenario 1, while the maximum power curtailment is 83 TWh in scenario 4. This demonstrates a deficit of the power supply in 2030 when encountered with extreme weather events. There is a strong need to prepare power supply plans for extreme events. Different from the results of 2030, both power shortages and curtailments are high in 2060 (Supplementary Fig. 5b). The finding reveals that there exist an

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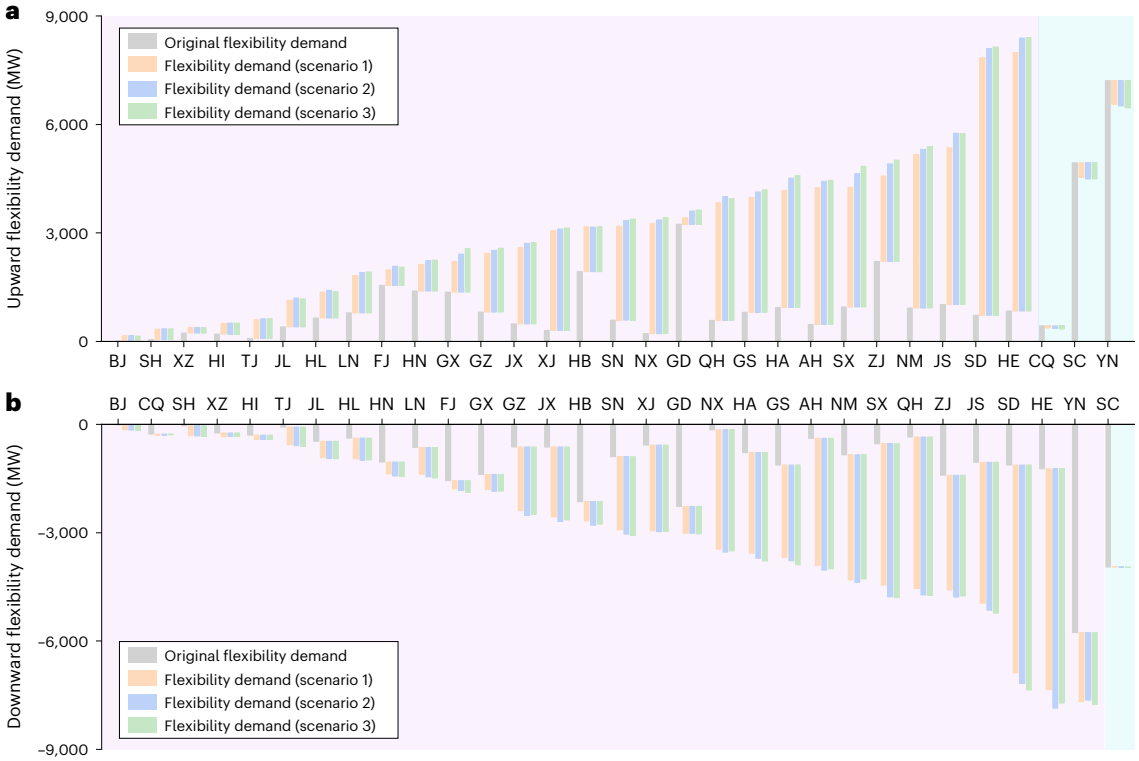


Fig. 5 | The flexibility demand in different scenarios. a, UFD (95th percentile). **b**, DFD (95th percentile). The timescale is 1 h. The pink area represents the provinces where the flexibility demand of net load is larger than that of the original load, while the blue represents the opposite.

uneven generation capability and lack of interprovincial power transmission channels. Among all scenarios, scenario 4 shows the smallest power shortage. This is because monthly renewable power generation (Supplementary Fig. 5d) is more even than in the other scenarios over the entire horizon. It is also implied that future climate change for the SSP126 is relatively favourable to power systems. By contrast, scenario 1 presents large power generations in April and May (Fig. 3e), thereby leading to significant power curtailments during this period. However, the power shortage peaks occur in December. Such uneven seasonal distribution might require additional constructions of seasonal energy storage. More details about scenario 1 at three stages can be found in Supplementary Tables 2–4.

Analysis of flexibility demand

The flexibility demand depends on fluctuations of the load and the intermittent renewable generation. Therefore, the net load that is calculated by deducting wind, solar, thermal, nuclear and receiving power from the original load is used to quantify the flexibility demand in this study. We analyse province-level flexibility demands because provincial power grids are responsible for the power balance. The upward flexibility demand (UFD) and downward flexibility demand (DFD) for different scenarios in the current stage are represented in Fig. 5a and Fig. 5b, respectively. The integration of VRE leads to a sharp increase in flexibility demands in most provinces, such as Jiangxi (JX), Ningxia (NX), NM, Shandong (SD) and HE. The main reason is that the trend of wind power profile is opposite to load demand, and the solar power profile sharply increases and decreases. Moreover, the increases are relatively high in northern provinces. One of the main reasons is the high heating loads and low solar power generation in winter. Such features result in the increasing power flexibility demand. Differing from northern China, the increases in flexibility demand in the southern provinces are relatively low owing to the consistent trend in load and VRE generation profiles. The temperature in summer is about 5–10 °C higher than the annual average temperature, especially in the afternoon, so the load demand has a summer afternoon peak driven by air-conditioning usage, which is consistent with the peak period of solar generation. Thus, there is a low degree of flexibility demand in these provinces.

More specifically, the average UFD and DFD of all the provinces for scenario 1 is 3,119 MW and 2,909 MW, and increased by 166% and 180% compared with the original values, respectively. Similar conclusions were found for other scenarios. These data indicate that the integration of VRE has a greater impact on the DFD. The average UFD and DFD are highest with 3,273 MW and 3,015 MW in scenario 3, respectively. This implies that the strong wind and solar radiation have the largest impact on flexibility demands.

We select six provinces (Jiangsu (JS), Zhejiang (ZJ), Anhui (AH), GD, GX and YN) to explore the characteristics of the flexibility demand at different timescales. The variation of flexibility demand with timescale in different scenarios in Supplementary Fig. 6 reveals three results. First, flexibility demand profiles changing with different timescales present two peaks, in GX and YN. Second, the impact of timescales on UFD and DFD has opposite patterns. For example, the peak of UFD in GD occurs when the timescale is within 0–12 h, while the peak of DFD appears within 12–24 h. Third, the flexibility demands under different scenarios are almost identical among small timescales. When the timescale is larger, the differences in flexibility demand under different scenarios become greater, such as JS and GX.

Supplementary Fig. 7 displays the distribution of flexibility demands for scenario 1 in 2023, by 2030 and by 2060, with a timescale of 1 h. Power flexibility demands are higher in the eastern and central regions, while lower in the west and northeast regions. The average values in 2030 and 2060 are 1.7 and 6.1 times that of the current stage, respectively. This implies that provincial power systems with high penetration of VRE will face increasing demand for dispatchable power. This phenomenon is particularly prominent in HE, SD and JS, where solar power accounts for a high proportion of the total installed capacity.

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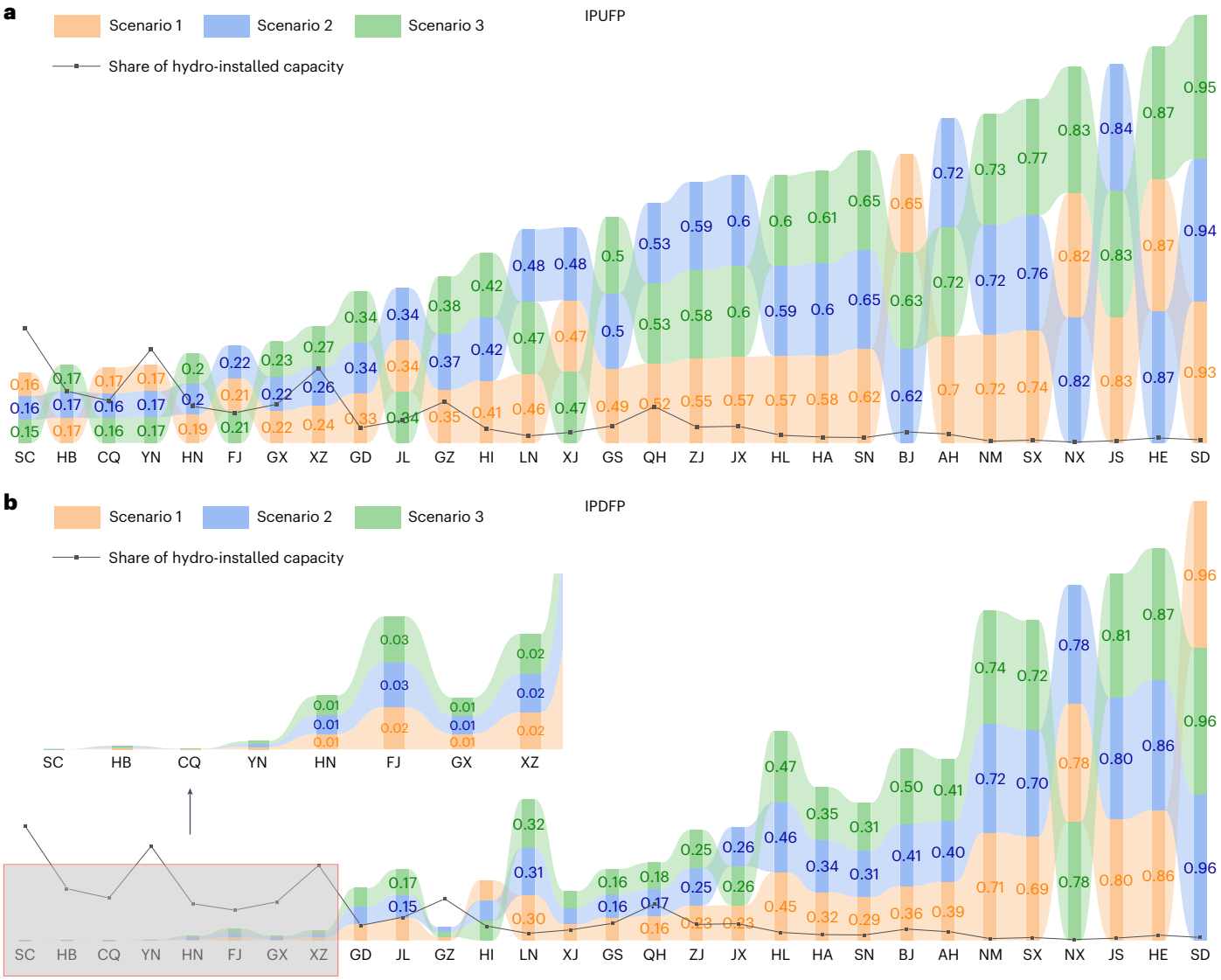


Fig. 6 | Insufficient probabilities of flexible power in 2023. a, IPUFP. **b**, IPDFP. The strip charts represent the IPUFP and IPDFP for each province.

Analysis of flexibility supply

This study focuses mainly on the flexibility supply of hydropower and energy storage in the hybrid system[28]. This section analyses the supply–demand relationship of each provincial hybrid system. Readers can refer to the 'Evaluation method for flexibility supply' section in the Methods for specific evaluation methods.

Figure 6a and Fig. 6b display the insufficient probability of upward flexible power (IPUFP) and insufficient probability of downward flexible power (IPDFP) without energy storage, respectively. Most provinces have the lowest IPUFP and IPDFP when the wind and solar radiation are both extremely weak (scenario 1). On the contrary, the IPUFP and IPDFP are highest when the wind and solar radiation are extremely strong (scenario 3). Meanwhile, the shortfall between flexible power supply and demand becomes larger, which might cause severe power shortages and curtailments. We also find that the IPUFP and IPDFP increase as the share of hydropower decreases (Fig. 6, black line). There are 23 provinces with an IPUFP over 30% and 14 provinces with an IPDFP over 30%. The averages of IPUFP and IPDFP of all provinces are 51.5% and 32.8% in scenario 1, with a gap of 18.7%. The results of other extreme weather scenarios are like scenario 1. Besides, the IPDFP is lower than the IPUFP in most provinces. This is mainly due to low hydropower generating capability in extremely dry years, affecting the range for upward regulation. In this case, the shortfall between upward flexible power (UFP) and UFD is higher than that between downward flexible power (DFP) and DFD.

More specifically, we take ZJ as an example and draw probability distribution maps of flexibility demand, flexibility supply and flexibility margin under different extreme weather scenarios (Supplementary Fig. 8). The shapes of the flexibility demand distribution in each row are similar, indicating a tiny gap of flexibility demands among different extreme weather scenarios. As the VRE penetration increases, the distribution of flexibility demand and flexibility supply gradually becomes flat. The IPUFP and IPDFP increase with large rates of around 20% and 28% (from 2023 to 2060), respectively.

The distribution maps of IPUFP and IPDFP in scenario 1 are shown in Supplementary Fig. 9. The provinces with high IPUFP and IPDFP are concentrated mainly in the northeast currently, mainly due to the high share of VRE and scarcity of flexible power supply. However, the scope will extend from northeast to southwest gradually as the VRE penetration increases in the future. By 2060, the averages of IPUFP and IPDFP will rise from 51.5% to 67.2% and from 32.8% to 52.4%, respectively.

In this context, the energy storage is further considered. The sizes of energy storage for all provinces under three extreme scenarios are shown in Supplementary Fig. 10. The power capacities of energy storage are close across different extreme scenarios in most provinces. Specially, the power capacity of energy storage under scenario 3 in JS

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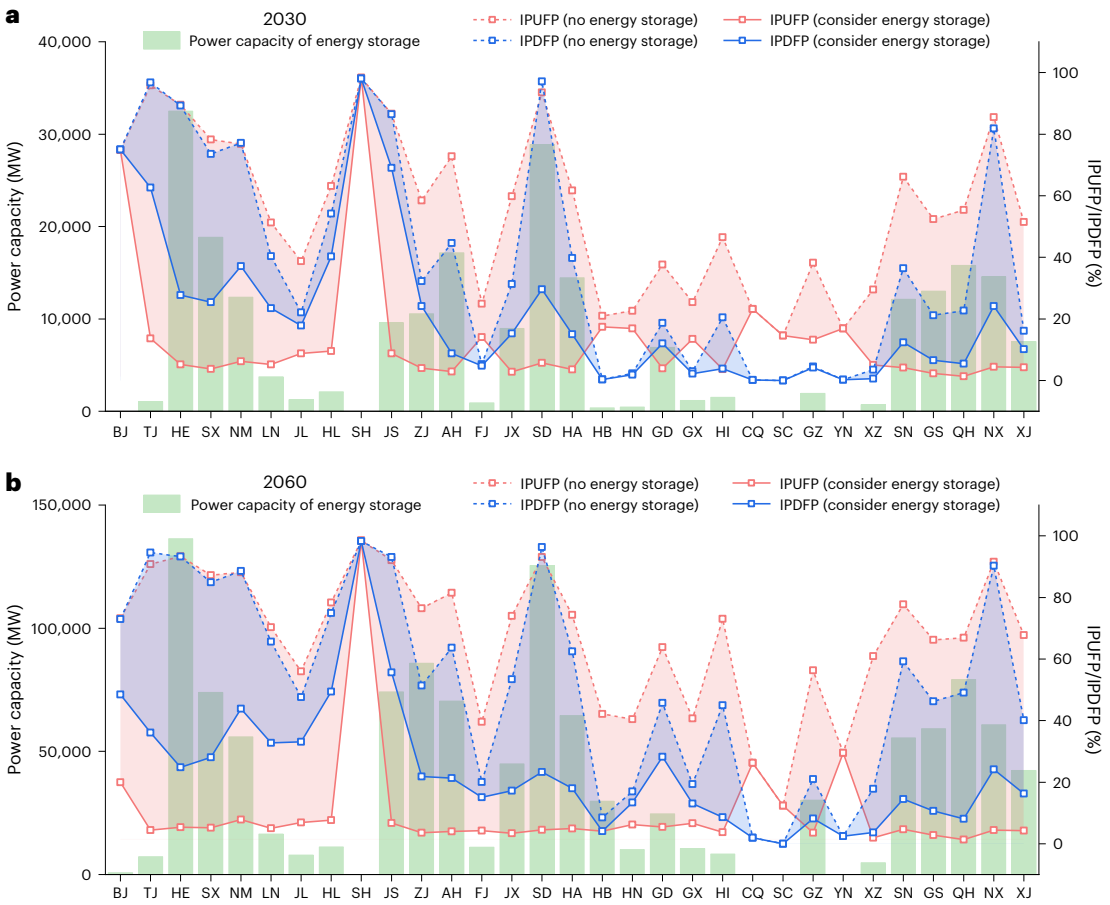


Fig. 7 | Power capacities of energy storage and IPUFP and UPDFP in scenario 1 under different stages. a, The power capacity of energy storage and IPUFP and IPDFP in scenario 1 in 2030. **b**, The power capacity of energy storage and IPUFP and IPDFP in scenario 1 in 2060.

in 2030 is much larger than under the other two scenarios. The reason is that the residual loads of most time periods in JS in 2030 are positive in scenarios 1 and 2 (Supplementary Fig. 11), meaning long-term power shortages. This is difficult to solve by installing large-scale energy storage and might require adjustments in the structure of power sources.

The discharge durations under different extreme scenarios have some differences in a few regions, such as Tianjin (TJ), Jilin (JL) and GZ in 2030 and NM, Liaoning (LN), Heilongjiang (HL), Fujian (FJ), JX and GZ in 2060. This is related to the duration of power shortages or curtailments. For example, the duration of power curtailments in TJ under scenarios 1 and 2 is much shorter than that under scenario 3 (Supplementary Fig. 12), resulting in shorter discharge duration needs. By contrast, results from scenarios 1 and 2 show that the power shortage in GZ has lasted for a longer period compared with those from scenario 3 (Supplementary Fig. 13). Correspondingly, longer discharge durations are required.

The sizes of energy storage for different provinces can be found in Supplementary Tables 5 and 6. This part focuses on the energy storage differences among provinces. The results of scenario 1 are used to analyse the storage capacity and the role of energy storage in enhancing the power system flexibility. Figure 7 shows the optimized power capacities of energy storage. The results reveal that provinces with a large intermittent renewable capacity commonly exhibit strong demand for energy storage capacity, such as HE, Shanxi (SX), NM, JS, ZJ, AH and SD. This is due to large and frequent generation fluctuations of wind and solar power. Through installing the energy storage, the flexible power supply of these provincial power grids has been greatly improved. In Fig. 7, a significant reduction can be seen for the IPUFP and IPDFP. Their averages reduce by 41.9% and 17.3% in 2030, respectively. The role of energy storage might become more significant in 2060, with larger reduction of 57.2% and 29.0% for IPUFP and IPDFP, respectively.

By contrast, other provinces with a low demand for energy storage fall into two categories. The first category includes FJ, Hubei (HB), Hunan (HN), GX, Chongqing (CQ), Sichuan (SC), YN and Tibet (XZ). They have large installed hydropower capacities and, thus, can provide much dispatchable power for the integration of wind and solar power. The second category contains Beijing (BJ) and Shanghai (SH), which show a whole power shortage in 2030 and 2060 (Supplementary Tables 3 and 4). In this situation, both regions lack excess electricity to charge energy storage devices. As a result, there is low demand for installing new energy storage.

The sizes of energy storage in 2060 and the growth rate from 2030 to 2060 under scenario 3 (this scenario presents the most unfavourable situation of insufficient flexibility) are shown in Supplementary Fig. 14. The sizes of energy storage in most provinces show a sharp increase from 2030 to 2060. This is because significant increases in installed capacities of wind and solar power cause large fluctuations of the net load. In particular, the growth rates have exceeded ten times those in FJ, HB, HN, GX and GZ. Such significant growth can be attributed to the uneven changes in supply and demand of flexible power in two stages. These provinces rely mainly on hydropower to provide dispatchable power. In the stage of 2030, hydropower can provide strong support for the power balance, so the additional demand for energy storage is relatively small. However, by 2060, the significant increase in wind and solar power capacity will lead to a significant gap in system flexibility. This change stimulates a strong demand for energy storage.

In addition, the results indicate that there exist severe seasonal power shortages in 2060 (Supplementary Table 4). It is difficult to

achieve electricity transfer over a long period of time by configuring short-duration energy storage, and this requires seasonal energy storage in many provinces. In terms of the monthly power shortage and curtailment, there are ten regions with seasonal storage demands, including BJ, TJ, HE, JX, SD, Henan (HA), HN, GD, Shaanxi (SN) and Xinjiang (XJ) (Supplementary Table 7).

Discussion and conclusions

The main purpose of this study is to evaluate the impact of extreme weather combinations on China's hybrid system. Through the analysis of inflow, wind and solar capacity factors over 40 years, representative extreme weather events are selected. We combine multiple extreme weather conditions into three unfavourable combinations and conduct a systematic analysis on the power generating capability, flexibility demand and supply under these weather scenarios.

The generating capability of the entire system exhibits four obvious characteristics and changes. First, the annual generating capability under extreme weather has a maximum decrease of 10.0% compared with the annual average value. The extreme drought leads to a reduction of 261 TWh in annual hydropower generation, while extremely strong wind and solar radiation might result in a power curtailment of 97 TWh. Second, owing to the uneven distribution of hydropower, wind and solar resources, the national generating capability is generally higher in the west and lower in the east. Third, with the renewable energy development from the current stage to 2030 and 2060, the generating peak period of the hybrid system will move from June and July to April and May, mainly due to the transition from a high-proportion hydropower system to a system dominated by wind and solar power. The annual utilization hours of the hybrid system under extreme drought, weak wind and solar radiation will decrease from 1,951 h (current stage) to 1,907 h (2030) and then down to 1,722 h (2060). Fourth, the interprovincial electricity exchange has a great effect on the power balance of provincial power grids and helps to decrease power shortages by 100%, 75.3% and 69.9%, respectively, in the current stage, 2030 and 2060. There is a need for additional interprovincial transmission channels in the future, especially between China Southern Power Grid and State Grid Corporation of China.

The power flexibility demand presents different spatiotemporal variation characteristics in the country. First, the extremely strong wind and solar radiation (scenario 3) have the largest impact on flexibility demands, 4.9% (UFD) and 3.7% (DFD) more than the value under extremely weak wind and solar radiation weathers (scenario 1). Second, the distribution of flexibility demand shows a trend of large in the eastern and central regions and small in the west and northeast regions, affected mainly by the load and VRE fluctuations.

More specifically, flexible power supply–demand relationships have been summarized depending on provincial power balance entities. First, when the wind and solar radiation are extremely strong, the insufficient probabilities of flexible power of most provincial power grids are largest with an average IPUFP of 52.5% and an average IPDFP of 34.2%. This is dependent on immense flexibility demand under the weather combination of extremely strong wind and strong solar radiation, as well as extreme drought. Second, the IPUFP and IPDFP in the northern regions are generally higher than those in the southern regions, with a gap of over 25% for the IPUFP and 29% for the IPDFP, respectively. Third, the flexibility supply provided by hydropower will be far from meeting power flexibility demand by 2060. The role of energy storage will be significant.

In summary, either insufficient generating capability or UFP supply will lead to electricity shortages. The combination of extreme drought, extremely weak wind and solar radiation might result in an annual electricity shortage of 424 TWh in 2030, accounting for 3.7% of the annual power consumption. In terms of UFP, the shortfall expectation between UFP and UFD could be up to about 36 GW in 2030 and will be expanded by more than four times in 2060 under the considered weather combinations, imposing tremendous pressure on supply power. Furthermore, insufficient DFP can result in an evident shortfall between DFP and DFD. If the combination of extreme drought and strong wind and solar radiation occurs, the shortfall expectation could reach 25 GW in 2030.

The above analysis and discussion have inspired the following suggestions.

First, with the continuous growth of wind, solar and hydropower capacity, the difficulty of power balance shows an upward trend under extreme weather events. It is essential to equip the power grid with a proper proportion of stable power sources that are less affected by meteorological factors to avoid large-scale extensive electricity shortages. The required stable power capacity should be dependent on the electricity gap.

Second, extreme drought with strong wind and solar radiation causes a large lack of flexible power. In a few regions with large installed hydropower capacity, such as southwest China, transforming conventional large hydropower plants into hybrid pumped-storage plants is a feasible way to increase the long-term power flexibility of the electrical system[29]. By comparison, in the other regions with little hydropower, there is a necessity to construct a certain scale of long duration and seasonal storage energy plants[30].

Methods

Description of extreme weather

Extreme weather indicators. In this Article, inflow, wind capacity factor and solar capacity factor are used to represent the level of hydropower, wind and solar resources, respectively.

Due to the consistent interannual variation in inflow for hydropower stations within the same basin, a typical hydropower station is selected to represent the hydropower resource of the basin where it is located. The historical annual inflows are sorted in descending order and the empirical frequency of each year's inflow is calculated using

$$\mathrm{Pr}_i = \frac{R(W_i)}{I} \tag{1}$$

where W_i is the inflow in the year i; I is total years; $R(W_i)$ is the ranking of inflow in the year i among all the years; and Pr_i is the empirical frequency of inflow in the year i.

Based on the empirical frequency obtained from the above equation, the 40 years are divided into five levels: extremely wet year ($0 \le \mathrm{Pr}_i < 20\%$), wet year ($20\% \le \mathrm{Pr}_i < 40\%$), normal year ($40\% \le \mathrm{Pr}_i < 60\%$), dry year ($60\% \le \mathrm{Pr}_i < 80\%$) and extremely dry year ($80\% \le \mathrm{Pr}_i \le 100\%$).

The reanalysis data of ERA5 are used as input data to calculate the wind and solar capacity factors. We mainly utilize the wind speed of 100 m, solar radiation intensity and temperature each hour, with a spatial resolution of 0.25° × 0.25°. The capacity factors of each province are calculated on the basis of the weighted average of the capacity factors of the grids it contains, and we obtain the current year's capacity factor by averaging the hourly capacity factors. Through the above method, we obtain a series of capacity factors for each province in 40 years.

The capacity factors calculated by reanalysis data are different from the actual capacity factors, because wind and solar power plants are not established in every power grid. The sites of wind and solar power plants are usually with abundant resources and far away from cities. In other words, the capacity factors calculated by the meteorological data are mainly used to reflect the resources abundance of every year and interannual changes in energy resources. Moreover, these capacity factors for renewables are further corrected on the basis of the actual capacity factors that are calculated using utilization hours from the National Energy Administration[31]. Thus, the course data provide useful support to capture the capacity factors of renewables.

Finally, the national capacity factors are obtained by calculating the weighted average of capacity factors of all provinces based on the installed capacity of wind power or solar power. The wind and solar

capacity factors are calculated by Supplementary Equation (1) and Supplementary Equation (2) in Supplementary Note 1.

Using the same standards as inflow, the abundance of wind and solar resources is divided into five levels (extremely weak, weak, normal, strong and extremely strong) based on the empirical frequency of wind and solar capacity factors.

Extreme weather scenarios. The current extreme weather scenarios adopt actual historical extreme scenarios, and the future extreme weather scenarios are obtained by multiplying historical scenarios with climate change factors.

In this study, the slope method is used to describe the climate change trends. First, the linear relationship between capacity factors and corresponding years is linearized using the least-squares method, as shown in equation (2). Second, according to the developed relationship, the capacity factors in 2030 and 2060 are calculated, and thus, the years with extreme weather can be obtained. Third, the ratio of capacity factors in future and in the years with extreme weathers is calculated and taken as climate change factors, as shown in equation (3). Finally, the future extreme scenarios are determined by multiplying the current extreme scenarios by the corresponding climate change factors.

$$\begin{aligned} y &= k^{\text{wind}}x + c^{\text{wind}} \\ y &= k^{\text{solar}}x + c^{\text{solar}} \end{aligned}, \tag{2}$$

where k^{wind}, c^{wind}, k^{solar} and c^{solar} are parameters of the linear function between capacity factors and years. x and y represent the year and capacity factor, respectively.

$$\eta = \frac{y_*}{y_\cdot}, \tag{3}$$

where η is the climate change factor, y_* denotes the capacity factor in 2030 or 2060 and $y_\cdot$ denotes the capacity factor in the year with a certain extreme weather condition, including extremely weak wind, extremely strong wind, extremely weak solar radiation and extremely strong solar radiation.

To further consider the impacts of future climate change, we have added two extreme scenarios from CMIP6, SSP126 and SSP585. We searched all available data for the whole Chinese region from CMIP6 in 2030 and 2060 and found a set of complete data with the highest precision, which is temporal (1 day) and spatial (100 km × 100 km) resolution for SSP126 and SSP585. To satisfy the demands for the present operation model, we further deal with these data by meteorological downscaling techniques. Two steps are contained. First, the bilinear interpolation is used to interpolate wind speed, radiation and temperature data from CMIP6 onto a 0.25° × 0.25° grid, respectively. Second, the curve decomposition is used to decompose the daily average wind speed (or radiation and temperature) to hourly data based on historical intraday wind speed (or radiation and temperature) trend.

Evaluation method for generating capability

Generating capability of different power. Based on the provincial hydropower–wind–solar renewable energy system, the evaluation methods for hydropower, wind power and solar power generating capability are proposed. The generating capability of wind and solar power is calculated by multiplying the capacity factor by the installed capacity for all provinces. The generating capability of hydropower is determined through an optimization model. Here, main hydropower plants in large rivers are considered in the model.

The objective of the model used in this study is to enhance the average generating capability of hydropower plants in the dry season and, thus, guarantee monthly electricity supply, especially dispatchable flexibility of electrical grids, over the entire time horizon. In actual hydropower operations, this objective is often used to optimize the long-term generation schedules of hydropower stations. Therefore, the objective function is formulated with maximizing the minimum monthly power generation, shown as

$$\begin{aligned} \max P &= \min_{m=1}^{M} P_{b,m}^{\text{hydro}} \\ P_{b,m}^{\text{hydro}} &= \sum_{n=1}^{N_b} P_{b,n,m}^{\text{hydro}}, \end{aligned} \tag{4}$$

where $P_{b,n,m}^{\text{hydro}}$ is the power generation of hydropower station n in basin b in month m, MW; $P_{b,m}^{\text{hydro}}$ is the sum of power generation of all hydropower stations in basin b in month m, MW; b and B are the index and total number of basins, respectively; n and N_b are the index and total number of hydropower stations in basin b, respectively; and m and M are the period index and total number of months over the entire time horizon, respectively.

The decision variables are monthly generation $P_{b,n,m}^{\text{hydro}}$ and discharge flow $Q_{b,n,m}^{\text{power}}$ of each hydropower station. This means that we can obtain the monthly generation profiles and discharge flow profiles of all hydropower stations by optimizing these decision variables. Certainly, the corresponding forebay level profiles can also be obtained.

The above objective function is subject to some important operation conditions and constraints, including generation production functions, continuity equation, outflow function, reservoir outflow boundaries, power discharge boundaries, reservoir storage volumes at the beginning and ending of time horizon, forebay elevation boundaries, power generation boundaries, relationship between forebay elevation and reservoir volume, and relationship between tailwater elevation and outflow (see Supplementary Note 2 for constraint details).

Due to the occurrence of spilled water before full capacity in individual months of the generating plan, it is necessary to adjust the generation of the months with spilled water. The correction strategy is

$$P_{b,n,m}^{\text{hydro}} = \min\left(A_{b,n} \min\left(Q_{b,n,m}^{\text{out}}, \text{QU}_{b,n,m}^{\text{power}}\right) H_{b,n,m}, \text{PU}_{b,n,m}^{\text{hydro}}\right). \tag{5}$$

The generating capability of hydropower in a certain province that is involved in the eight major basins in this Article is calculated by equation (6), while others are estimated by the actual utilization hours in the considered year, as shown in equation (7). The national generating capability of hydropower is calculated by equation (8).

$$P_{s,m}^{\text{hydro}} = \frac{\text{PZ}_s^{\text{hydro}}}{\text{PZ}_s^{\text{hydro}} - \text{PZ}_s^{\text{hydro-a}}} P_{s,m}^{\text{hydro-a}}, \tag{6}$$

where $P_{s,m}^{\text{hydro}}$ is the hydropower generation in month m of province s, MW; $P_{s,m}^{\text{hydro-a}}$ is the power generation in month m of province s obtained through the above method, MW; $\text{PZ}_s^{\text{hydro}}$ is the hydropower installed capacity of province s, MW; and $\text{PZ}_s^{\text{hydro-a}}$ is the hydropower installed capacity of provinces s included in the eight major basins, MW.

$$P_{s,m}^{\text{hydro}} = \frac{T_{s,m}^{\text{hydro}}}{\Delta t_m} \text{PZ}_s^{\text{hydro}}, \tag{7}$$

where $T_{s,m}^{\text{hydro}}$ is the equivalent full load utilization hours of hydropower in month m of province s.

$$P_m^{\text{hydro}} = \sum_{s=1}^{S} P_{s,m}^{\text{hydro}}, \tag{8}$$

where P_m^{hydro} is the national hydropower generation in month m, MW.

The monthly generating capability of wind power is determined by

$$\begin{aligned} P_{s,m}^{\text{wind}} &= \text{CWP}_{s,m}^{\text{month}} \text{PZ}_s^{\text{wind}} \\ \text{CWP}_{s,m}^{\text{month}} &= \sum_{t \in \lambda_m} \sum_{k \in \varphi_s} \text{CWA}_{s,k,t}^{\text{hour}} \gamma_{s,k} / \Delta t_m \\ P_m^{\text{wind}} &= \sum_{s=1}^{S} P_{s,m}^{\text{wind}} \end{aligned}, \tag{9}$$

where $CWP_{s,m}^{month}$ is the wind capacity factor in month m of province s; $P_{s,m}^{wind}$ is the wind power generation in month m of province s, MW; P_m^{wind} is the national wind power generation in month m, MW; and λ_m is the periods included in month m.

The monthly generating capability of solar power is determined by

$$\begin{aligned} P_{s,m}^{solar} &= CSP_{s,m}^{month} PZ_s^{solar} \\ CSP_{s,m}^{month} &= \sum_{t\in\lambda_m}\sum_{k\in\varphi_s} CSA_{s,k,t}^{hour}\gamma_{s,k}/\Delta t_m \\ P_m^{solar} &= \sum_{s=1}^{S} P_{s,m}^{solar} \end{aligned}, \quad (10)$$

where $CSP_{s,m}^{month}$ is the solar capacity factor in month m of province s; $P_{s,m}^{solar}$ is the solar power generation in month m of province s, MW; and P_m^{solar} is the national solar power generation in month m, MW.

For each provincial power grid, the installed capacities of various types of power sources at the current stage, 2030 and 2060 are originated from actual data and expanded data of China's 2030 energy and electricity development plan and outlook for 2060, respectively.

Interprovincial power exchange model. Provincial power grids are the core dispatching institute in China. They are responsible for the whole power balance between demand and supply. Here, we consider main interprovincial power transmission channels among different provincial power grids and develop an interprovincial power exchange model. This model aims to optimize the power exchange among provincial power grids. The objective is to minimize the national power shortage within the entire time horizon. It should be noted that the consideration of interprovincial power transmission channels might require a monthly redistribution of thermal energy production to be suitable for renewable power generation with seasonable characteristics. Therefore, at this stage, the monthly thermal power generations are still optimized. The decision variables are the monthly transmitting electricity and thermal power generation of each province. The main dispatching constraints include power balance, the calculation of power exchange, power transmission capacity limitation, the calculation of monthly hydropower, wind power, solar power and nuclear power generation, and thermal power generation constraint. The model is shown in equation (11).

$$\begin{aligned} &\min \sum_{m=1}^{M}\left(\sum_{s=1}^{S} E_s^{short}\right)^2 \\ &s.t.\ E_{s,m} = E_{s,m}^{thermal} + E_{s,m}^{nuclear} + E_{s,m}^{hydro} + E_{s,m}^{wind} + E_{s,m}^{solar} \\ &\qquad + E_{s,m}^{netreceive} + E_{s,m}^{short} - E_{s,m}^{surplus} \\ &E_{s,m}^{netreceive} = E_{s,m}^{receive} - E_{s,m}^{send} \\ &E_{s,m}^{receive} = \sum_{sr=1}^{S} TR_{sr,s} E_{sr,s,m}^{trans}\rho \\ &E_{s,m}^{send} = \sum_{sr=1}^{S} TR_{s,sr} E_{s,sr,m}^{trans} \\ &E_{s,sr,m}^{trans} \le PU_{s,sr}^{trans} \times \Delta t_m \\ &E_{s,m}^{hydro} = P_{s,m}^{hydro}\Delta t_m \\ &E_{s,m}^{wind} = P_{s,m}^{wind}\Delta t_m \\ &E_{s,m}^{solar} = P_{s,m}^{solar}\Delta t_m \\ &E_{s,m}^{nuclear} = PZ_s^{nuclear} T_{s,m}^{nuclear} \\ &\sum_{m=1}^{M} E_{s,m}^{thermal} = \sum_{m=1}^{M} PZ_s^{thermal} T_{s,m}^{thermal} \end{aligned}, \quad (11)$$

where *s.t.* represents the constraints of the model. $E_{s,m}$ is the consumption in month m of province s, MWh. $E_{s,m}^{thermal}$, $E_{s,m}^{nuclear}$, $E_{s,m}^{hydro}$, $E_{s,m}^{wind}$ and $E_{s,m}^{solar}$ are the generation of thermal, nuclear, hydropower, wind power and solar power in month m of province s, respectively, MWh. It is worth noting that the thermal power referred here includes coal fired power, gas power, biomass power and hydrogen generation. $E_{s,m}^{short}$ and $E_{s,m}^{surplus}$ are the power shortage or power surplus in month m of province s, MWh. $E_{s,m}^{netreceive}$ is the net receiving power of province s in month m, MWh. $E_{s,m}^{netreceive} > 0$ represents receiving power greater than sending power, while $E_{s,m}^{netreceive} < 0$represents receiving power less than sending power. $E_{s,m}^{send}$ and $E_{s,m}^{receive}$ are the sending and receiving power of province s in month m, respectively, MWh. $E_{sr,s,m}^{trans}$ is the electricity transmitted from province sr to province s in month m, MWh. ρ represents the line loss rate. $TR_{sr,s}$ denotes the transmission channel between province sr and province s. $TR_{sr,s} = 1$means that province sr can transmit electricity to province s. On the contrary, if it is equal to 0, this transmission direction will be unusable. If $TR_{sr,s} = 0$ for all provinces, it implies that interprovincial power interactions are not considered. $PU_{s,sr}^{trans}$ is the channel capacity between province sr and province s, MW. Δt_m denotes the number of hours in month m, h. $T_{s,m}^{thermal}$ and $T_{s,m}^{nuclear}$ are the equivalent full load utilization hours of thermal and nuclear power in month m of province s, respectively; $PZ_s^{thermal}$ and $PZ_s^{nuclear}$ are the installed capacities of thermal and nuclear power in province s, respectively, MW.

It is important to note that this study mainly presents long-term electricity exchange and balance analysis for the national power system, with a large time step. In real-time operation, owing to the potential complex constraints faced by the power grid and power stations, the power shortage and surplus might exceed the calculated results in the study.

Evaluation method for flexibility demand

In this Article, the provincial power grids are taken as the study object when analysing flexibility demand and supply. The flexibility demand is quantified by the fluctuation of the net load, as shown in equation (12).

$$\begin{aligned} \Delta L_{s,t}^{\tau} &= L_{s,t}^{net} - L_{s,t-\tau}^{net} \\ L_{s,t}^{net} &= L_{s,t} - P_{s,t}^{wind-hour} - P_{s,t}^{solar-hour} - P_{s,t}^{thermal-hour} \\ &\quad - P_{s,t}^{nuclear-hour} - P_{s,t}^{netreceive-hour} \end{aligned}, \quad (12)$$

where $L_{s,t}^{net}$ and $L_{s,t-\tau}^{net}$ are net loads of province s at period t and $t-\tau$, respectively, MW. $\Delta L_{s,t}^{\tau}$ is the fluctuation of province s at period t when the timescale is τ, MW. $L_{s,t}$ is the load of province s at period t, MW; τ is the timescale, which can be taken as 1 h, 2 h,... 24 h in this study. $P_{s,t}^{wind-hour}$, $P_{s,t}^{solar-hour}$, $P_{s,t}^{thermal-hour}$ and $P_{s,t}^{nuclear-hour}$ are the wind, solar, thermal and nuclear power generation of province s at period t, respectively, MW. $P_{s,t}^{netreceive-hour}$ is the net receiving power of province s at period t, MW. $P_{s,t}^{netreceive-hour} > 0$represents receiving power from other provinces, while $P_{s,t}^{netreceive-hour} < 0$ represents sending power to other provinces. They are calculated by equation (13).

$$\begin{aligned} P_{s,t}^{wind-hour} &= CWP_{s,t}^{hour} PZ_s^{wind} \\ CWP_{s,m}^{hour} &= \sum_{k\in\varphi_s} CWA_{s,k,t}^{hour}\gamma_{s,k} \\ P_{s,t}^{solar-hour} &= CSP_{s,t}^{hour} PZ_s^{solar} \\ CSP_{s,t}^{hour} &= \sum_{k\in\varphi_s} CSA_{s,k,t}^{hour}\gamma_{s,k} \\ P_{s,t}^{thermal-hour} &= E_{s,m}^{thermal}\frac{L_{s,t}}{\sum_{t\in\lambda_m} L_{s,t}} \\ P_{s,t}^{nuclear-hour} &= E_{s,m}^{nuclear}/\Delta t_m \\ P_{s,t}^{netreceive-hour} &= E_{s,m}^{netreceive}\frac{L_{s,t}}{\sum_{t\in\lambda_m} L_{s,t}} \end{aligned}, \quad (13)$$

where $CWP_{s,t}^{hour}$ and $CSP_{s,t}^{hour}$ are the wind and solar capacity factor at period t of province s, respectively.

The loads for 2023 are derived from actual values or typical daily extension data of provincial power grids, and the loads for 2030 and 2060 are calculated on the basis of the loads for 2023 and the growth rate forecasted by Global Energy Interconnection Development and Cooperation Organization[9].

Furthermore, flexibility demands can be divided into UFD and DFD, as shown in

$$\begin{aligned}\phi_s^{up} &= \left\{\Delta L_{s,t}^{\tau} | \Delta L_{s,t}^{\tau} > 0\right\} \\ \phi_s^{down} &= \left\{\Delta L_{s,t}^{\tau} | \Delta L_{s,t}^{\tau} < 0\right\}\end{aligned}, \tag{14}$$

where ϕ_s^{up} and ϕ_s^{down} are the sets of UFD and DFD of provinces s.

The cumulative distribution functions (CDFs) of UFD and DFD are shown in

$$\begin{aligned}Gup_s(x) &= P(\Delta L \le x), \Delta L \in \phi_s^{up} \\ Gdown_s(x) &= P(\Delta L \le x), \Delta L \in \phi_s^{down}\end{aligned}, \tag{15}$$

where $Gup_s(x)$ and $Gdown_s(x)$ are the CDFs of UFD and DFD, respectively.

In this Article, the 95th percentile is adopted to evaluate the flexibility demand of the provincial power grid. The equations are as follows:

$$\begin{aligned}D_s^{up} &= Gup_s^{-1}(0.95) \\ D_s^{down} &= Gdown_s^{-1}(0.05),\end{aligned} \tag{16}$$

where D_s^{up} and D_s^{down} are the UFD and DFD of provinces s, respectively, MW.

Evaluation method for flexibility supply

The flexible power supply of a provincial power grid equipped with energy storage can be calculated by equation (17).

$$\begin{aligned}C_{s,t}^{UR} &= \min\left(N_s^{ramp}, PU_{s,t}^{hydro} - P_{s,t}^{hydro-hour}\right) \\ &\quad + \min\left(\left(E_{s,t}^{storage} - EU_s^{storage}(1-DOD)\right)\zeta/1\,h, PU_s^{storage}\right) \\ C_{s,t}^{DR} &= \min\left(N_s^{ramp}, P_{s,t}^{hydro-hour} - PL_{s,t}^{hydro}\right) \\ &\quad + \min\left(\left(EU_s^{storage}DOD - E_{s,t}^{storage}\right)/\zeta/1\,h, PU_s^{storage}\right) \\ PU_{s,t}^{hydro} &= \max_{t\in\lambda_{m,d}} P_{s,t}^{hydro-hour}\end{aligned}, \tag{17}$$

where $C_{s,t}^{UR}$ and $C_{s,t}^{DR}$ are the UFP and DFP at period t of province s, respectively, MW; $P_{s,t}^{hydro-hour}$ is the hydropower generation at period t of province s, MW; N_s^{ramp} is the ramping ability of hydropower in province s, MW; $PU_{s,t}^{hydro}$ is the upper limit of hydropower generation at period t of province s, MW; $E_{s,t}^{storage}$ is the remaining electricity at period t of province s, MWh; $PU_s^{storage}$ is the power capacity for energy storage of province s, MW; $EU_s^{storage}$ is the energy capacity of province s, MWh; DOD is the depth of discharge; ζ is the round-trip efficiency; and $\lambda_{m,d}$ is the periods included in day d of month m. The time step is 1 h.

The hourly hydropower generation profiles for all provincial power grids throughout a year are optimized by 8,760 h production simulation. The provincial power grids are taken as the study object, and the objective is to minimize the sum of squared residual loads. The main dispatching constraints includes monthly hydropower generation constraint, hydropower regulating capability constraint and hourly hydropower generation boundary. The model is shown in equation (18).

$$\begin{aligned}&\min \sum_{t=1}^{T} L_{s,t}^{re^2} \\ &s.t. L_{s,t}^{re} = L_{s,t}^{net} - P_{s,t}^{hydro-hour} \\ &\sum_{t\in\lambda_{m,d}} P_{s,t}^{hydro-hour}/24 = P_{s,m}^{hydro} \\ &P_{s,m,d}^{hydro-day-max} = \max_{t\in\lambda_{m,d}} P_{s,t}^{hydro-hour} \\ &P_{s,m,d}^{hydro-day-min} = \min_{t\in\lambda_{m,d}} P_{s,t}^{hydro-hour} \\ &P_{s,m,d}^{hydro-day-max} \le PEAK_s P_{s,m,d}^{hydro-day-min} \\ &P_{s,t}^{hydro-hour} \le PU_s^{hydro}\end{aligned}, \tag{18}$$

where $L_{s,t}^{re}$ is the residual load of province s at period t, MW; $\lambda_{m,d}$ are the periods included in day d of month m; $P_{s,m,d}^{hydro-day-max}$ and $P_{s,m,d}^{hydro-day-min}$ are the maximum and minimum output of hydropower in day d of month m, respectively, MW; $PEAK_s$ is the coefficient of peak regulation of hydropower in province s; and PU_s^{hydro} is the upper limit of hydropower generation of province s, MW.

The residual load profile of each province by coordinating hourly generations of various power sources without the consideration of energy storage is obtained through the above model. Moreover, we can clearly know during which periods each province experiences power shortages and curtailments. Then, the obtained results can be used in the following energy storage sizing optimization model.

The power capacity and duration of the energy storage are optimized through an energy storage planning model. The charging/discharging profiles of dispatching all energy storages during the entire time horizon are also obtained. The objective is to minimize the economic loss of power shortage and the cost of energy storage system. The main dispatching constraints include power balance equation, energy storage transition, upper and lower limits of stored electricity, charge and discharge power limit, and initial and final state of energy storage. The model is shown in equation (19).

$$\begin{aligned}&\min \sum_{t=1}^{T} \vartheta P_{s,t}^{short} \times 1\,h + PU_s^{storage} DT_s^{storage} \varpi/\Delta y \\ &s.t. L_{s,t}^{re} = P_{s,t}^{discharge} - P_{s,t}^{charge} + P_{s,t}^{short} - P_{s,t}^{curtail} \\ &E_{s,t+1}^{storage} = E_{s,t}^{storage} + P_{s,t}^{charge}\zeta \times 1\,h - P_{s,t}^{discharge}/\zeta \times 1\,h \\ &E_{s,t}^{storage} \le PU_s^{storage} DT_s^{storage} DOD \\ &E_{s,t}^{storage} \ge PU_s^{storage} DT_s^{storage}(1-DOD) \\ &P_{s,t}^{charge} \le PU_s^{storage} BIN_{s,t} \\ &P_{s,t}^{discharge} \le PU_s^{storage}(1-BIN_{s,t}) \\ &E_{s,1}^{storage} = E_{start}^{storage} \\ &E_{s,T+1}^{storage} = E_{end}^{storage}\end{aligned}, \tag{19}$$

where ϑ is the penalty factors for power shortage, US$ per MWh. ϖ is the total installed cost of energy storage system related to the duration, US$ per MWh. Δy is the life span of the energy storage system. $P_{s,t}^{short}$ and $P_{s,t}^{curtail}$ are the power shortage and curtailment at period t of province s, respectively, MW. $P_{s,t}^{discharge}$ and $P_{s,t}^{charge}$ are the discharge and charge power at period t of province s, respectively, MW. $BIN_{s,t}$ is a binary variable. $BIN_{s,t} = 1$ represents charge, and $BIN_{s,t} = 0$ represents discharge. $E_{s,t}^{storage}$ is the remaining electricity at period t of province s, respectively, MWh. $DT_s^{storage}$ is the duration of the energy storage and can only be taken as one of the integers 2, 4, 6, 8 or 10 considering current energy storage technology. $E_{start}^{storage}$ and $E_{end}^{storage}$ are the initial and final stored electricity of energy storage, respectively, MWh.

Further, we determine discrete probability distributions of flexibility demand and flexibility supply through statistics methods. The probability distribution of flexibility margin is obtained by calculating the convolution difference between probability distributions of flexible power supply and demand. The equations are formulated as follows:

$$\begin{aligned}\kappa up_s(z) &= fup_s(y)\Theta gup_s(x)\\ \kappa down_s(z) &= fdown_s(y)\Theta gdown_s(x),\end{aligned} \tag{20}$$

where κup_s and $\kappa down_s$ are probability density functions (PDFs) of upward flexibility margin and downward flexibility margin, respectively; gup_s and $gdown_s$ are the PDFs of UFD and DFD, respectively; fup_s and $fdown_s$ are PDFs of UFP and DFP, respectively; and Θ denotes the convolution difference.

This study uses the IPUFP and IPDFP to evaluate the flexibility of provincial power grids, shown as follows:

$$\begin{aligned}IPUFP_s &= \int_{-\infty}^{0}\kappa up_s(z)dz\\ IPDFP_s &= \int_{-\infty}^{0}\kappa down_s(z)dz.\end{aligned} \tag{21}$$

When flexible power is insufficient, the upward and downward power gaps are respectively calculated by equation (22).

$$\begin{aligned}SUF_s &= \int_{-\infty}^{0} z\times\kappa up_s(z)dz\\ SDF_s &= \int_{-\infty}^{0} z\times\kappa down_s(z)dz\end{aligned}, \tag{22}$$

where SUF_s is the shortfall expectation between UFP and UFD of province s, MW; and SDF_s is the shortfall expectation between DFP and DFD of province s, MW.

Data availability

ERA5 data are available at https://www.ecmwf.int/. CMIP6 data are available at https://aims2.llnl.gov/search. The operating parameters and run-off data for reservoirs and hourly load cannot be disclosed owing to our non-disclosure agreements with power grid corporations. However, they can be obtained from the corresponding author upon reasonable request. Other data used in this Article are provided in the Supplementary Information (Supplementary Tables 8–18).

Code availability

The optimal scheduling model for hydropower is solved by Lingo 18. The interprovincial power exchange model, 8,760 h production simulation and energy storage sizing optimization model are implemented via Python 3.10 and solved by Gurobi 10.0.1. The evaluation method for flexibility demand and supply is implemented via MATLAB R2018a. All the core codes are in the Supplementary Information. Other codes can be obtained from the authors upon reasonable request.

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Author contributions

J.S., Y.W. and M.L. designed and performed the experiments. J.S. and Y.W. analysed the data and wrote the paper. C.C., J.K.K., J.W., X.G., X.L., B.Z. and L.G. reviewed and edited the paper.

Competing interests

The authors declare no competing interests.

Additional information

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